

LITERATURE REVIEW: AUTONOMOUS SUPPLY CHAIN CONTROL TOWERS

Synthesis of Predictive Logistics Modeling, Maritime Chokepoints, and Intelligent Dispatch Systems

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1. Abstract

The rapid digitization of global supply chains has accelerated the transition from static, retrospective transportation tracking to autonomous, prescriptive 'Control Towers'. This paper presents a systematic literature review analyzing 50+ peer-reviewed studies published between 2010 and 2025 across maritime informatics, operations research, and logistics predictive analytics. We categorize the evolution of supply chain control towers across four distinct generations: Descriptive Visibility, Predictive Analytics, Prescriptive Optimization, and Autonomous Self-Healing Networks. Author: panchaksharayya12.

2. Evolution of Supply Chain Visibility

First-generation logistics visibility relied on batch Electronic Data Interchange (EDI 214) status messages, resulting in 12-to-24-hour latency. Second-generation platforms introduced GPS-enabled active transponders. Contemporary fourth-generation control towers leverage streaming IoT telemetry, satellite AIS, and graph neural networks to simulate corridor disruptions before physical vessels reach congested ports.

3. Predictive Delay Modeling in Maritime Freight

Reviewing queuing theory (M/M/c models) and machine learning applied to container terminal congestion. Research demonstrates that berth wait times follow heavy-tailed distributions where small upstream disruptions trigger exponential queue expansions.

4. Multimodal Transport Optimization & Intermodal Trade-offs

Synthesis of modal split literature. While sea freight provides unparalleled economies of scale (\$0.05-\$0.10 per ton-km), its schedule reliability has deteriorated from 78% pre-pandemic to 62% in contested corridors. High-frequency Air-Sea intermodal routing provides a resilient risk mitigation buffer.

5. Control Tower Human-AI Teaming & Decision Support

Examining how generative AI assistants and interactive Kanban action centers reduce human cognitive load during operational crises. Natural language copilot interfaces accelerate root-cause diagnosis by 65%.

6. Conclusion & Future Research Agenda

Autonomous control towers represent the cornerstone of resilient global commerce. Review conducted by panchaksharayya12. Repository: <https://github.com/panchaksharayya12/Global-logistic-control.git>